

Spotlight on Technology Case Study

AI Chips

In 1999 semiconductor maker Nvidia made history when it introduced the graphics processing unit (GPU), which is used for handling complex images and large quantities of data for graphics-related tasks in video games and other software. It turns out that Nvidia's GPUs also work really well for machine learning and generative AI applications, which must be trained using massive quantities of data in order to recognize intricate patterns, sounds, pictures, and text and to create chatbots and generate images. As a result, Nvidia is making history again as the world's leading producer of the chips powering AI applications. Nvidia-specialized AI chips underpin most of today's AI applications. It has been said that Nvidia controls as much as 80 percent of the global market for specialized AI chips. Recent generative AI applications from Microsoft, OpenAI, Amazon, and Meta could not be built without Nvidia's hardware. For instance, ChatGPT was trained using 10,000 of Nvidia's GPUs clustered together in a supercomputer belonging to Microsoft.

A GPU differs from the computer's central processing unit (CPU) in how it processes data and the number of physical processing units (cores) it uses. A CPU only has a few cores and performs serial computing – processing data one task after another. It handles all the main functions of a computer and is designed to handle a wide range of tasks quickly. A GPU has hundreds of cores and performs parallel computing, processing many calculations at once. It is more specialized for handling high-resolution images and videos concurrently. Its high throughput and ability to perform many calculations with large quantities of data simultaneously make it ideal for training large AI models.

None of the Big Tech companies like depending so heavily on Nvidia, nor does the U.S. government. The United States currently produces less than 10 percent of the world's chips and none of the most advanced logic chips, some of which power AI. The United States designs advanced logic chips, but its inability to manufacture them domestically has emerged as a national security and economic risk. The U.S. CHIPS and Science Act, or CHIPS Act, enacted in 2022, dedicates \$39 billion in subsidies for creating or expanding U.S.-based semiconductor fabrication and an additional \$11 billion in research and development. Leading recipients of CHIPS Act funds include Intel, Samsung, Micron Technology, and GlobalFoundries. The Biden administration anticipates that CHIPS investments for Intel and other semiconductor manufacturers will enable the United States to produce roughly 20 percent of the world's most advanced logic chips by 2030, up from the current level of zero. Producing more semiconductors domestically will help insulate U.S. chip supplies from major supply chain disruptions and dependence on Taiwan and other Asian countries that currently dominate global chip production.

The Big Tech firms are all starting to invest heavily in designing their own custom chips, which would support larger AI models and give them more control over their AI businesses. Having better computer chips could soon become a bigger competitive advantage than having better computer code. AI chips made in-house could be more easily tailored to a company's particular AI models, making its products more efficient and energy-conserving. If an iPhone, Google Pixel, or Microsoft Surface tablet can run a more powerful generative AI model and load its results a bit faster because of a custom microchip, then more customers might buy that device. That could be a game-changer.

Google has been able to train and run its flagship Gemini models with less energy and at lower cost by using custom computer processors rather than relying on Nvidia. However, many Google Cloud servers also run on Nvidia chips, and Google optimized its Gemma language model to run on Nvidia GPUs. Amazon touts its custom AI chips for their superior machine learning training performance at

significantly lower costs. But the firm is also growing its partnership with Nvidia. Microsoft believes that its custom chips enhance its systems rather than replacing existing hardware powered by Nvidia.

Every company wants its own self-contained IT platform, no longer beholden to competitors' prices or external supply chain disruptions. But whether any of these cloud computing tech companies can rival or replace Nvidia remains an open issue right now, and it is highly unlikely that any of them will sever ties with Nvidia. It may well be that the dominant pattern for the future is a hybrid combination of custom computer chips and Nvidia designs.

Nvidia is currently working on even more powerful chips capable of training next-generation AI models for generative AI applications such as ChatGPT, recommender systems, and data analytics. In March 2024, CEO Jensen Huang announced a new generation of high-performance AI chips called Blackwell, which can reduce energy consumption for training large AI models by a factor of 3 to 4 compared to the previous generation of GPUs. Blackwell has 208 billion transistors and is an upgrade of the company's H100 chip, which is currently the most advanced GPU in production. Blackwell is 30 times quicker at some tasks than its predecessor. Blackwell enables connecting many GPUs so that large AI models can be trained with a smaller carbon footprint. Among the many organizations expected to adopt Blackwell are Amazon Web Services, Dell Technologies, Google, Meta, Microsoft, OpenAI, Oracle, Tesla, and xAI.

CASE STUDY QUESTIONS

Why are Nvidia chips so useful for AI applications?

Nvidia has a near-monopoly on specialized AI chips. Discuss the impact of this situation.

Should companies develop their own custom AI chips or use those produced by Nvidia? Explain your answer.